

Arithmetic Structure of the Standard Model Mass Spectrum and the Bost-Connes System

Shao-Wu Hu

Independent Researcher, Shanghai, China

forest@xindong.com ORCID: 0009-0009-8300-4691

Abstract

The 12 fermion masses of the Standard Model span 12 orders of magnitude and have hitherto been treated as independent experimental inputs. Starting from the Bost-Connes system, this paper shows that the CRT decomposition of the character group $(\mathbb{Z}/7\mathbb{Z})^*$ uniquely determines six fermion sectors and three generations, and provides a unified mass formula reducing all 12 masses to four Dirichlet L -function values at the critical line $s = 1/2$. All nine charged fermion predictions agree with experiment to within $\leq 2\%$. The neutrino masses $m_1 = 5.113$, $m_2 = 10.01$, $m_3 = 50.67$ meV and mass ratio $R = 34.31$ are new predictions of this work, with m_1 and m_2 not yet directly measured.

Contents

1	Introduction	2
2	Characters and Particle Species	2
2.1	Complexity Hierarchy	3
2.2	Uniqueness Theorem	3
3	Three Generations, Color, Charge, and Spin	5
3.1	Three-Generation Structure	5
3.2	Color	7
3.3	Charge Quantization	7
3.4	Discrete Spin and Fermion Statistics	8
3.5	Fermion Quantum Number Summary	9
4	Jacobi Phase, Cabibbo Angle, and CKM	9
4.1	Cabibbo Angle	11
4.2	Wolfenstein Hierarchy as Doublet Hop-Count Expansion	11
4.3	Origin of $A = \sqrt{2/3}$	12

5	Mass Spectrum	13
5.1	Character Paths	14
5.2	Vertex Factors	16
5.3	Unified Mass Formula	17
5.4	Lepton Masses	17
5.5	d-Type Quark Masses	18
5.6	u-Type Quark Masses	19
5.7	Neutrino Masses	20
6	Discussion and Predictions	21
6.1	Summary Table	22
6.2	Testable Predictions	22
6.3	Open Problems	23
6.4	Conclusion	23

1 Introduction

The fermion mass spectrum of the Standard Model possesses a striking structure. The 12 fermions—three generations of charged leptons (e, μ, τ), three neutrinos (ν_1, ν_2, ν_3), and six quarks (u, d, c, s, t, b)—have masses spanning 12 orders of magnitude, from 173 GeV for the top quark to ~ 0.05 eV for neutrinos. All three generations carry identical quantum numbers; the sole distinction is mass. Inter-generational mixing is described by the CKM matrix (quarks) and the PMNS matrix (neutrinos), with the Cabibbo angle $\lambda \approx 0.225$ and the CP-violating phase δ_{CP} as key parameters. These masses and mixing parameters—at least 19 free parameters in total—are experimental inputs in the Standard Model; the theory itself does not predict their values [6, 22].

Yet these parameters are not mutually independent. Lepton masses satisfy the Koide formula [3] ($Q = 2/3$ to 0.03% accuracy); quark masses exhibit the Georgi–Jarlskog relation [4] ($m_b/m_\tau \approx 3$); CKM matrix elements decrease as powers of λ (the Wolfenstein hierarchy [5]). These empirical regularities suggest a unified algebraic structure underlying the mass spectrum.

This paper demonstrates that this structure can be found precisely in the Bost–Connes (BC) system [1, 2, 12]. The BC system

$$\mathcal{A} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*$$

is a quantum statistical mechanical system whose automorphism group $\widehat{\mathbb{Z}}^*$ encodes the arithmetic symmetry at every prime. At the prime $p = 7$, the local factor

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

carries both the weak-force direction ($\mathbb{Z}/2\mathbb{Z}$) and the three-generation cycle ($\mathbb{Z}/3\mathbb{Z}$) through its CRT decomposition. This paper will prove:

- The discrete quantum information of fermions (three generations, color, charge, spin) is precisely described by the character structure and CRT decomposition of $(\mathbb{Z}/7\mathbb{Z})^*$ (§3);
- CKM mixing parameters arise from the phase of Jacobi sums and the Hecke three-cycle, with deviations $< 2.5\%$ (§4);
- All 12 fermion masses are given by the unified formula

$$m_f = v \times |L|^{\text{pow}} \times \text{vertex} \times \text{Gen} \quad (1)$$

where L denotes Dirichlet L -function values at the critical line $s = 1/2$, with all nine charged fermion deviations $\leq 2\%$ (§5);

- Neutrino masses $m_1 = 5.113$, $m_2 = 10.01$, $m_3 = 50.67$ meV are testable new predictions (§5.6).

2 Characters and Particle Species

At the prime $p = 7$, the multiplicative group

$$(\mathbb{Z}/7\mathbb{Z})^* = \{1, 2, 3, 4, 5, 6\}$$

is a cyclic group of order six. Taking 3 as a primitive root,

$$(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}.$$

By the Chinese Remainder Theorem,

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

This is the starting point of the entire paper: a minimal finite group carrying both parity-type bisection and generation-type trisection.

The dual group $(\widehat{\mathbb{Z}/7\mathbb{Z}})^*$ likewise has six characters, denoted χ_n , $n = 0, 1, \dots, 5$, satisfying

$$\chi_n(3^k) = e^{2\pi i n k / 6}.$$

The orders of the characters are

$$1, 6, 3, 2, 3, 6.$$

- $(\mathbb{Z}/7\mathbb{Z})^*$ is a cyclic group of order six; its CRT decomposition provides two directions: $\mathbb{Z}/2\mathbb{Z}$ distinguishes symmetry types (real vs. complex), while $\mathbb{Z}/3\mathbb{Z}$ encodes the cyclic structure of internal degrees of freedom.
- The order of a character measures the depth of its coupling to the multiplicative group—higher order probes more internal structure.
- The order distribution $\{1, 6, 3, 2, 3, 6\}$ naturally falls into four tiers, with the physics of each tier determined by the order itself.

2.1 Complexity Hierarchy

We interpret “order” as complexity level: higher order means a longer multiplicative period before returning to the identity, hence a more complex, more strongly coupled sector. This yields the following stratification:

Character order	Complexity	Particle sector	Description
1	Simplest	Photon/trivial gauge sector	Trivial vacuum representation
2	Real bisection	Neutrino	Self-conjugate, Majorana tendency
3	Three-cycle	Lepton	Phase trisection structure
6	Maximal complexity	Quark	Carries both $\mathbb{Z}/2$ and $\mathbb{Z}/3$

Here “photon” is not one of the six fermion types but rather labels the massless trivial gauge sector corresponding to the trivial character; the six nontrivial fermionic degrees of freedom are given by the remaining character orbits and conjugate pairings. More precisely, particle species are not “assigned” but uniquely selected by the following four axioms.

2.2 Uniqueness Theorem

Definition 2.1 (Four Classification Axioms). We seek to organize six group elements or six characters into the six fermion sectors of the Standard Model, subject to:

- (1) **Conjugation closure**: each physical sector must be closed under the involution $a \mapsto 7 - a$;
- (2) **Three-generation compatibility**: sector groupings must be compatible with $\mathbb{Z}/3\mathbb{Z}$ orbits;
- (3) **Complexity monotonicity**: character order must correspond monotonically to physical complexity: order 2 is simpler than order 3, which is simpler than order 6;

- (4) **Real/complex phase separation:** the unique real nontrivial character (Legendre character) must correspond to a self-conjugate sector and not be mixed with complex cubic characters.

Remark 2.2 (Why “higher order = more complex”?). Axiom (3) is not an arbitrary assumption but a natural consequence of the KMS states and L -function structure of the BC system:

- **Order 1** (χ_0 , trivial character): completely immune to KMS phase transitions; the BC state is trivial at all temperatures $\rightarrow m = 0$ (gauge protection!) $\rightarrow$ physically simplest;
- **Order 2** (χ_3 , Legendre character): real character, Gauss sum purely imaginary $\rightarrow$ tree-level $m = 0 \rightarrow$ next simplest;
- **Order 3** (χ_2, χ_4): complex characters, nontrivial Jacobi phase, mass determined by phase structure $\rightarrow$ intermediate complexity;
- **Order 6** (χ_1, χ_5): contains all CRT directions, richest KMS state structure $\rightarrow$ most complex.

Complexity monotonicity is therefore not an added assumption but a natural consequence of the KMS states [27] and Euler product structure of L -functions in the BC system.

On the surface, there are $6! = 720$ ways to label six objects into six categories; but these 720 “ambiguities” are completely eliminated by the four axioms above.

Theorem 2.3 (Uniqueness Theorem). *Under the four axioms above, the assignment of the six characters of $(\mathbb{Z}/7\mathbb{Z})^*$ to the six fermion sectors of the Standard Model is unique up to renaming isomorphism. Specifically: the unique order-2 real character must correspond to the neutrino sector; the unique pair of conjugate order-3 characters must correspond to the lepton sector; the two pairs of order-6 characters correspond to the two quark sectors. All 720 permutations collapse to a single physical solution under the axiomatic constraints.*

Proof. First consider the order structure. Among the six characters, there is exactly one trivial character of order 1; exactly one nontrivial real character of order 2; one conjugate pair of order-3 characters; and two pairs of order-6 characters. By axiom (3), the three complexity tiers—order 2, 3, 6—are thus uniquely fixed.

Next, axiom (4): the only nontrivial real character is the Legendre character, which is self-conjugate and can only correspond to a sector that is its own antiparticle; among fermions, this uniquely matches the Majorana-type neutrino sector. Order 2 therefore uniquely falls on the neutrino.

For the order-3 tier, there is only one conjugate pair, and its phases close on cube roots of unity, maximally compatible with the $\mathbb{Z}/3\mathbb{Z}$ three-generation period; assigning it to quarks would strip quarks of the maximal complexity of carrying both $\mathbb{Z}/2$ and $\mathbb{Z}/3$, violating axiom (3). Hence order 3 uniquely corresponds to the lepton sector.

The remaining two pairs of order-6 characters automatically correspond to the quark sectors. Since axiom (1) demands conjugation closure and axiom (2) demands compatibility with three-generation orbits, any attempt to swap “lepton/quark” or “neutrino/lepton” would violate order structure, self-conjugacy, or three-generation phase compatibility. The classification is therefore unique under automorphisms and physical relabeling. $\square$

The assignment of particle species is thus uniquely determined by the character structure of $p = 7$. The following table summarizes all mappings:

Character	Order	Conjugate pair	Fermion sector	Determining axiom
χ_0	1	—	Gauge vacuum	—
χ_3	2	Self-conjugate	Neutrino	(3)(4)
χ_2, χ_4	3	One pair	Charged lepton	(1)(2)(3)
χ_1, χ_5	6	Two pairs	Quark (u-type/d-type)	(1)(2)(3)

3 Three Generations, Color, Charge, and Spin

The fundamental particles of the Standard Model carry a set of discrete quantum numbers: three-generation structure ($e/\mu/\tau$ and corresponding quarks and neutrinos), particle–antiparticle conjugation (C transformation), color charge (three colors for quarks, colorless for leptons), charge quantization ($0, \pm 1/3, \pm 2/3, \pm 1$), weak isospin ($T_3 = \pm 1/2$), and spin ($J = 1/2$) together with fermion/boson statistics. These quantum numbers completely determine the identity of each fermion.

This section proves that all the above quantum information has precise arithmetic counterparts in the local factor $(\mathbb{Z}/7\mathbb{Z})^*$ of the BC system: three generations correspond to three orbits of the involution, color corresponds to the resolution in the $\mathbb{Z}/3\mathbb{Z}$ direction of the CRT decomposition, charge corresponds to the multiplicative decomposition of characters, and spin corresponds to the half-order property of the generator.

3.1 Three-Generation Structure

Fermions are grouped by generation: first generation (e, ν_e, u, d), second generation (μ, ν_μ, c, s), third generation (τ, ν_τ, t, b). Fermions within the same generation are tightly linked by gauge interactions; across generations, quantum numbers are identical—the only distinction is mass: $m_e = 0.511$ MeV vs $m_\tau = 1777$ MeV (leptons), $m_u \approx 2$ MeV vs $m_t = 173$ GeV (quarks), with mass hierarchies spanning several orders of magnitude. Within each generation, particles and antiparticles are exchanged by C transformation (charge conjugation), a $\mathbb{Z}/2\mathbb{Z}$ symmetry satisfying $C^2 = \text{id}$.

In the BC framework, particle–antiparticle conjugation corresponds to the involution on $(\mathbb{Z}/7\mathbb{Z})^*$:

$$\iota(a) = 7 - a, \quad \iota^2 = \text{id}.$$

Grouping the six elements by ι into pairs yields three doublets:

$$D_1 = \{1, 6\}, \quad D_2 = \{2, 5\}, \quad D_3 = \{3, 4\}.$$

The two elements $\{a, 7 - a\}$ in each doublet are exchanged by ι , corresponding to the particle–antiparticle pairing. The three doublets correspond to three generations.

Proposition 3.1 (Doublets and Three Generations). *The set $\{D_1, D_2, D_3\}$ comprises all orbits of $(\mathbb{Z}/7\mathbb{Z})^*$ under the involution $\iota(a) = 7 - a$. The particle–antiparticle pairing structure of three-generation fermions is precisely described by these three orbits.*

Although the three generations share identical quantum numbers, their masses differ enormously. The “distance” between the two members of each particle–antiparticle pair varies across generations: the first generation has the greatest internal asymmetry (lightest, most stable), while the third has the least (heaviest, fastest decaying). This inter-generational mass hierarchy reflects the degree of asymmetry within each particle–antiparticle pair.

In the BC framework, this asymmetry is described by the **tension** of a doublet. For a doublet $D = \{a, 7 - a\}$, define

$$T(D) := |a - (7 - a)| = |2a - 7|.$$

$T(D)$ measures the distance between the two members of the doublet: greater tension means greater asymmetry, tighter binding, and lower energy. Direct computation for the three doublets gives

$$T(D_1) = |1 - 6| = 5, \quad T(D_2) = |2 - 5| = 3, \quad T(D_3) = |3 - 4| = 1.$$

The tension sequence is therefore

$$\{5, 3, 1\}.$$

$T(D_1) = 5$ (largest) corresponds to the lightest first generation; $T(D_3) = 1$ (smallest, members nearly coincide) corresponds to the heaviest third generation. The ordering of tensions precisely mirrors the physical mass hierarchy: this pattern holds simultaneously for charged leptons, quarks, and neutrinos—all three fermion types share the same generational structure, with the mass hierarchy ordering determined by a single set of doublets.

Furthermore, the tension sequence 5, 3, 1 has a regular recursive structure—the internal asymmetry decreases by a fixed step across each generation. In the BC framework, this regularity is precisely captured by the arithmetic progression of tensions:

Theorem 3.2 (Tension Arithmetic Theorem). *For the three involution doublets of $p = 7$,*

$$D_1 = \{1, 6\}, \quad D_2 = \{2, 5\}, \quad D_3 = \{3, 4\},$$

the tension sequence satisfies

$$T(D_1), T(D_2), T(D_3) = 5, 3, 1,$$

forming an arithmetic progression with common difference 2. The common difference $2 = d_1$, the order of the $\mathbb{Z}/2\mathbb{Z}$ factor in the CRT decomposition.

Proof. For $D_k = \{k, 7 - k\}$,

$$T(D_k) = |k - (7 - k)| = |2k - 7|.$$

Substituting $k = 1, 2, 3$ gives 5, 3, 1. Consecutive differences are all 2, hence an arithmetic progression with common difference 2. $\square$

The tension decreases by the fixed step $d_1 = 2$, yet the mass hierarchy itself is far from uniform ($m_\tau/m_\mu \approx 17$, $m_\mu/m_e \approx 207$). This enormous disparity arises from the nonlinear mapping of tensions to masses via Koide rotation, to be given in §5.

The three doublets $D_1 = \{1, 6\}$, $D_2 = \{2, 5\}$, $D_3 = \{3, 4\}$ are given by $\mathbb{Z}/3\mathbb{Z}$ Galois orbits. Each doublet corresponds to one fermion generation. Different fermion types (leptons, quarks, neutrinos) within the same generation are distinguished by character order (§3.2), while the mass assignments among three generations of the same type will be determined by the Koide formula in §5.

3.2 Color

Quarks carry three color charges (red, green, blue), while leptons carry no color. In the BC framework, the threefold nature of color can be described using the three positions $a_2 \in \{0, 1, 2\}$ in the $\mathbb{Z}/3\mathbb{Z}$ direction of the CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Whether a character can distinguish these three positions is determined by its order.

Definition 3.3 (Kernel and Color). The kernel of character χ_j is $\ker \chi_j = \{a \in (\mathbb{Z}/7\mathbb{Z})^* : \chi_j(a) = 1\}$, with $|\ker \chi_j| = 6/\text{ord}(\chi_j)$. Physically, quarks participate in strong interactions (carry color charge) while leptons do not (colorless). In the BC framework, this distinction is determined by character order:

- $\text{ord} = 6$ (χ_1, χ_5): $|\ker| = 1$. Probes both the $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ directions, fully resolving all 6 elements. Corresponds to color-carrying quarks, with $n_c = d_2 = 3$ color channels.
- $\text{ord} = 3$ (χ_2, χ_4): $|\ker| = 2$. Blind to the $\mathbb{Z}/2\mathbb{Z}$ direction. Corresponds to colorless charged leptons, $n_c = 1$.
- $\text{ord} = 2$ (χ_3): $|\ker| = 3$. Blind to the $\mathbb{Z}/3\mathbb{Z}$ direction, completely color-blind. Corresponds to colorless neutrinos, $n_c = 1$.

The kernel size $|\ker| = 6/\text{ord}$ measures a character's “resolution” of the internal structure of $(\mathbb{Z}/7\mathbb{Z})^*$. Larger $|\ker|$ means more group elements are folded into the same phase, appearing colorless externally. Quarks carry color while leptons do not; in the BC framework, this distinction is described by character order: colored characters ($\text{ord} = 6$, $|\ker| = 1$) carry complete $(\mathbb{Z}/7\mathbb{Z})^*$ information, with the three color channels $d_2 = 3$ corresponding to the three cosets of $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\zeta_7)^+)$; colorless characters ($\text{ord} \leq 3$) lack sufficient phase information to distinguish conjugate pairs, giving only one effective channel.

3.3 Charge Quantization

Electric charge is quantized: leptons carry integer charges (0 or -1), quarks carry fractional charges ($+2/3$ or $-1/3$). In the BC framework, electromagnetic interaction is described by the order-3 character χ_2 . The charge Q measures the coupling strength of each sector in the χ_2 direction, with the coupling of colored sectors shared equally among $n_c = d_2 = 3$ color channels.

Using the character multiplication $\chi_i \cdot \chi_j = \chi_{i+j \bmod 6}$ and $\bar{\chi}_2 = \chi_4$, the charges of each fermion type have natural arithmetic interpretations in the BC framework:

- Charged leptons (e, μ, τ) have charge $Q = -1$. In BC, charged leptons correspond to χ_2 , the EM character itself, with coupling strength 1.
- Neutrinos (ν) have charge $Q = 0$. In BC, neutrinos correspond to χ_3 , which contains no χ_2 component and does not couple to the electromagnetic force.
- d-type quarks (d, s, b) have charge $Q = -1/3$. In BC, $\chi_5 = \chi_2 \cdot \chi_3$ contains a χ_2 component, but quarks carry color, so the EM coupling is shared among $d_2 = 3$ color channels, giving $Q = -1/d_2 = -1/3$.
- u-type quarks (u, c, t) have charge $Q = +2/3$. In BC, $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains $\bar{\chi}_2$, likewise shared among $d_2 = 3$ color channels. In the weak isospin doublet (u, d) , $T_3(u) = +1/2$, $T_3(d) = -1/2$. From $Q(d) = -1/3$ the hypercharge is $Y = Q - T_3 = -1/3 + 1/2 = +1/6$; the same doublet shares the hypercharge, so $Q(u) = T_3 + Y = 1/2 + 1/6 = +2/3$.

Theorem 3.4 (Charge Formula).

$$Q(e, \mu, \tau) = -1 \quad (\text{pure } \chi_2) \quad (2)$$

$$Q(\nu) = 0 \quad (\text{pure } \chi_3, \text{ no } \chi_2 \text{ component}) \quad (3)$$

$$Q(d, s, b) = -\frac{1}{d_2} = -\frac{1}{3} \quad (\chi_2 \text{ shared by color}) \quad (4)$$

$$Q(u, c, t) = T_3 + Y = +\frac{2}{3} \quad (\text{doublet structure}) \quad (5)$$

3.4 Discrete Spin and Fermion Statistics

Fermions carry spin $1/2$ (sign change under half-turn, return to identity under full turn), and fermions and bosons obey different statistics. In the BC framework, these properties arise naturally from the doublet structure.

Theorem 3.5 (Discrete Spin). *The defining property of spin $1/2$ is: rotation by half a turn (π) changes the sign of the state vector, while a full turn (2π) restores it. In the BC framework, this property is described by the generator $g = 3$ (order 6) of $(\mathbb{Z}/7\mathbb{Z})^*$. By Euler's criterion,*

$$g^{(p-1)/2} = 3^3 = 27 \equiv -1 \pmod{7},$$

i.e., the “half-turn” (g^3) of the generator equals -1 . On the antisymmetric representation of the doublet $\{a, 7-a\}$, -1 acts as $-\text{id}$:

$$g^3|\psi\rangle = -|\psi\rangle \quad (\text{half-turn sign change}), \quad g^6|\psi\rangle = +|\psi\rangle \quad (\text{full-turn restoration}).$$

This precisely reproduces the physical behavior of spin $1/2$.

Proof. $3^3 = 27 = 4 \times 7 - 1 \equiv -1 \pmod{7}$, a direct consequence of Euler's criterion. The involution $\iota: a \mapsto 7-a$ on the doublet is precisely the action of multiplication by -1 . The antisymmetric state satisfies $\iota|\psi\rangle = -|\psi\rangle$ (half-turn sign change); $\iota^2 = \text{id}$ gives full-turn restoration. $\square$

Theorem 3.6 (Fermion/Boson $\mathbb{Z}/2\mathbb{Z}$ Classification). *Fermions and bosons obey different quantum statistics: fermion states change sign under particle exchange (antisymmetric), while boson states are unchanged (symmetric). In the BC framework, this classification is given by the eigenvalue decomposition of the involution ι on doublets:*

- *Symmetric state ($\iota = +1$): $a + (7-a) = 7$ (constant, no information)—corresponds to boson statistics;*
- *Antisymmetric state ($\iota = -1$): $a - (7-a) = 2a - 7 = \pm T$ (tension, carries information)—corresponds to fermion statistics.*

The Legendre character χ_3 provides the arithmetic criterion for this classification: $\chi_3(\iota) = +1$ (QR) corresponds to bosons; $\chi_3(\iota) = -1$ (QNR) corresponds to fermions.

Remark 3.7. Conventionally, spin $1/2$ is defined via continuous rotations of $\text{SU}(2)$ ($\pi_1(\text{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$). In the present framework, spin is given directly by the $\mathbb{Z}/2\mathbb{Z}$ subgroup $\{1, -1\}$ of $(\mathbb{Z}/7\mathbb{Z})^*$, without introducing continuous groups. The two descriptions are equivalent in their physical predictions.

Remark 3.8 (Half-integer spin and quadratic extension). $-1 \equiv 6 \pmod{7}$ is not a quadratic residue ($\sqrt{-1}$ does not exist in $\mathbb{F}_7$) [15]. Half-integer spin requires the quadratic extension $\mathbb{F}_{7^2}$, corresponding to the double cover $\text{SO}(3) \rightarrow \text{SU}(2)$. This is the arithmetic origin of the spin-statistics theorem.

3.5 Fermion Quantum Number Summary

From §2 (characters and orders) and §3 (CRT decomposition, doublets, and involution), all quantum numbers of each fermion type have now been determined:

Sector	χ_j	Order	Q	T_3	$SU(3)_c$	J	Particles
u-type quark	χ_1	6	$+2/3$	$+1/2$	3	$1/2$	u, c, t
d-type quark	χ_5	6	$-1/3$	$-1/2$	3	$1/2$	d, s, b
Charged lepton	χ_2	3	-1	$-1/2$	1	$1/2$	e, μ, τ
Neutrino	χ_3	2	0	$+1/2$	1	$1/2$	ν_1, ν_2, ν_3

All discrete quantum information of fermions—three-generation structure, color charge, charge quantization, weak isospin, spin, and fermion/boson statistics—finds precise arithmetic counterparts in the BC framework. Three generations arise from three orbits of the involution, color from the resolution in the $\mathbb{Z}/3\mathbb{Z}$ direction of the CRT decomposition, charge quantization from the multiplicative decomposition of characters and sharing among color channels, and spin from the half-order property of the generator. These correspondences involve no continuous parameters or fitting—they are completely determined by the arithmetic structure of $(\mathbb{Z}/7\mathbb{Z})^*$. The remaining continuous parameters—masses—will be given by L -function values at the critical line in §5.

4 Jacobi Phase, Cabibbo Angle, and CKM

Inter-generational mixing of quarks is among the most precisely measured phenomena in particle physics. Experiments show that mass eigenstates and weak interaction eigenstates do not coincide; the rotation between them is described by the CKM matrix [20, 19].

The CKM matrix is a 3×3 unitary matrix whose largest off-diagonal element—the Cabibbo angle $\lambda \approx 0.225$ —controls the transition amplitude between adjacent generations; transitions across more generations decrease as powers of λ , forming the clear Wolfenstein hierarchy [5]: $V_{us} \sim \lambda$, $V_{cb} \sim A\lambda^2$, $V_{ub} \sim \lambda^3$. Additionally, the CKM matrix contains a CP-violating phase, leading to matter–antimatter asymmetry.

This section proves that all CKM parameters have arithmetic counterparts in the BC framework. The mixing angles come from the Jacobi sum phase δ_2 , the cyclic structure of inter-generational transitions from the Hecke three-cycle, and the Wolfenstein hierarchy from the hop-count expansion of Hecke paths.

Cyclic structure of inter-generational transitions. Physically, the three generations of quarks (u, c, t) and (d, s, b) undergo weak decays forming a cycle: $u \rightarrow c \rightarrow t \rightarrow u$, $d \rightarrow s \rightarrow b \rightarrow d$. Each transition is accompanied by the emission or absorption of a W boson, and the entire cycle closes among three generations.

In the BC framework, this cyclic transition can be described by the Hecke-type action of multiplication by 2 on $(\mathbb{Z}/7\mathbb{Z})^*$. Direct computation of its action on the three doublets:

$$2 \cdot \{1, 6\} = \{2, 12\} \equiv \{2, 5\} = D_2,$$

$$2 \cdot \{2, 5\} = \{4, 10\} \equiv \{4, 3\} = D_3,$$

$$2 \cdot \{3, 4\} = \{6, 8\} \equiv \{6, 1\} = D_1.$$

Therefore

$$D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1,$$

and the Hecke action induces a three-cycle on doublet space, precisely corresponding to the physical inter-generational transition cycle.

Theorem 4.1 (Hecke Three-Cycle). *Let S_m^* denote the Hecke-type action on the BC local factor. At $m = 2$, S_2^* induces a three-cycle on the doublet set $\{D_1, D_2, D_3\}$:*

$$D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1.$$

In particular, its minimal positive order is 3.

Proof. The direct computation above gives $S_2^*(D_1) = D_2$, $S_2^*(D_2) = D_3$, $S_2^*(D_3) = D_1$. Therefore $S_2^{*3} = \text{id}$. Since neither S_2^* nor S_2^{*2} is the identity, the minimal positive order is 3. $\square$

The diagonal elements of the CKM matrix correspond to the amplitude of remaining in the same doublet, while off-diagonal elements correspond to the amplitude of hopping between doublets. Quark mixing is thus embedded in a discrete three-cycle path, rather than being an arbitrary 3×3 unitary matrix.

Mixing angle: Jacobi phase δ_2 . The *cyclic structure* of inter-generational transitions is determined by the Hecke three-cycle; the *angle* of transitions requires another quantity—the phase of the Jacobi sum.

Physically, the off-diagonal elements of the CKM matrix measure the mixing angle between adjacent generations. In the BC framework, this angle is given by the phase of the Jacobi sum $J(\chi_2, \chi_2) = \tau(\chi_2)^2 / \tau(\bar{\chi}_2)$ [15, 26]. The Jacobi sum measures the multiplicative autocorrelation of character χ_2 , and its phase $\arg(J)$ encodes the relative rotation angle among the three-generation doublets. The Eisenstein decomposition gives the exact value:

$$J(\chi_2, \chi_2) = -1 - d_2 \omega_3, \quad \omega_3 = e^{2\pi i/3},$$

where $d_2 = 3$. Thus $\arg J = -\arctan(d_2^{3/2}) = -\arctan(3\sqrt{3}) = -79.11^\circ$, giving

$$\delta_2 = \frac{\arg J(\chi_2, \chi_2)}{\varphi(p)} = \frac{-\arctan(d_2^{3/2})}{2d_2} = -13.18^\circ. \quad (6)$$

Modulus-phase separation: the absolute scale $|L(\frac{1}{2}, \chi_2)|^4$ is evaluated at the functional equation symmetry axis $s = \frac{1}{2}$; the Koide phase δ is evaluated at $s^* = \frac{1}{2} + Q/(2\pi)$, where $Q = 2/3$ is the Koide ratio (an algebraic identity uniquely determined by $r = \sqrt{d_1}$). Physically, the Koide rotation has angular step $2\pi/3$ in character space, corresponding to $\Delta s = (2\pi/3)/(2\pi) = 1/(3\pi)$ on the L -function spectral parameter axis, a fractional offset after normalizing by a full 2π period. This gives

$$\delta = \arg L\left(\frac{1}{2} + \frac{1}{3\pi}, \chi_2\right) = -12.73^\circ.$$

4.1 Cabibbo Angle

Experimentally, the Cabibbo angle $\lambda \approx 0.225$ is the largest off-diagonal CKM element, directly measuring the mixing strength between first- and second-generation quarks ($d \leftrightarrow s$ transition). In the BC framework, λ is uniquely determined by the Jacobi phase δ_2 : the single-step doublet transition amplitude is

$$\lambda = \frac{\sin(2\delta_2)}{2}.$$

Substituting $\delta_2 \approx 13.18^\circ$,

$$\lambda \approx 0.222,$$

agreeing with the experimental value $\lambda_{\text{exp}} \approx 0.2245$ to the percent level.

Proposition 4.2 (Single-Step Tunneling). *If the minimal inter-doublet transition is induced by the phase difference $2\delta_2$, then the single-step tunneling amplitude satisfies*

$$\lambda = \frac{\sin(2\delta_2)}{2} = 0.222,$$

agreeing with the experimental Cabibbo parameter $\lambda_{\text{exp}} \approx 0.2245$ to the percent level.

Remark 4.3 (Why $\sin(2\delta_2)/2$?). The functional form $\lambda = \sin(2\delta_2)/2$ has a precise character-theoretic interpretation. Since $\chi_5 = \chi_2 \times \chi_3$ and $\chi_1 = \bar{\chi}_2 \times \chi_3$, the χ_3 factor cancels in CKM matrix elements; thus CKM mixing arises entirely from interference between χ_2 ($\arg = -\delta_2$) and $\bar{\chi}_2$ ($\arg = +\delta_2$):

- $\sin(\delta_2)$: projection amplitude of the u-type sector in the mixing direction;
- $\cos(\delta_2)$: projection amplitude of the d-type sector in the mixing direction;
- Product $\sin(\delta_2) \cos(\delta_2) = \frac{1}{2} \sin(2\delta_2)$: superposition of the two projections, i.e., the CKM off-diagonal amplitude.

The factor $1/2$ comes from a trigonometric identity, not a free parameter.

4.2 Wolfenstein Hierarchy as Doublet Hop-Count Expansion

The magnitudes of CKM matrix elements exhibit a clear hierarchy: transitions between adjacent generations (V_{us}) are strongest, those skipping one generation (V_{cb}) are next, and those spanning two generations (V_{ub}) are weakest. The Wolfenstein parametrization [5, 20] expresses this hierarchy as powers of λ :

$$V_{us} \sim \lambda, \quad V_{cb} \sim A\lambda^2, \quad V_{ub} \sim \lambda^3(\rho - i\eta).$$

Physically, the exponent of λ corresponds directly to generational distance: adjacent generations $\rightarrow$ 1 step, skip one $\rightarrow$ 2 steps, skip two (full cycle) $\rightarrow$ 3 steps.

In the BC framework, the Wolfenstein hierarchy can be precisely described by the path length of the Hecke three-cycle: V_{us} corresponds to a single-step doublet hop, V_{cb} to a two-step process, V_{ub} to a complete three-step closed loop. A three-step closed loop starting from any doublet returns to itself; there are three equivalent paths, with the normalization coefficient of the Fourier projection $P_0 = (1 + T_2 + T_2^2)/3$ giving the factor $1/3$:

$$|V_{ub}| \sim \frac{\lambda^3}{3}.$$

Remark 4.4 (Precise origin of the $1/3$ factor). A three-step closed loop starting from any doublet returns to itself via three paths:

$$D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_1, \quad D_2 \rightarrow D_3 \rightarrow D_1 \rightarrow D_2, \quad D_3 \rightarrow D_1 \rightarrow D_2 \rightarrow D_3.$$

Each path contributes amplitude λ^3 but carries the Hecke three-cycle phase ω^k ($\omega = e^{2\pi i/3}$, $k = 0, 1, 2$). Coherent superposition:

$$\lambda^3(1 + \omega + \omega^2) = 0.$$

Complete cancellation. However, the closed-loop condition selects the $k = 0$ trivial Fourier mode (returning to the starting point = trivial representation), whose projection normalization is $1/|\mathbb{Z}/3\mathbb{Z}| = 1/3$:

$$|V_{ub}| = \lambda^3 \times \frac{1}{|\mathbb{Z}/3\mathbb{Z}|} = \frac{\lambda^3}{3}.$$

The $1/3$ is therefore the normalization coefficient of the Fourier projection $P_0 = (1 + T_2 + T_2^2)/3$, not a probability average.

Theorem 4.5 (Wolfenstein Hop-Count Law). *In the Hecke three-cycle model, the Wolfenstein hierarchy can be interpreted as a doublet hop-count expansion:*

$$V_{us} \sim \lambda^1, \quad V_{cb} \sim A\lambda^2, \quad V_{ub} \sim \frac{\lambda^3}{3}.$$

The exponents are precisely the shortest Hecke path lengths.

Explanation. A single adjacent hop gives the minimal suppression λ ; the shortest path for a two-step process has length 2, giving suppression λ^2 ; the three-step process corresponds to a full closed loop with 3 paths; averaging yields $\lambda^3/3$. A rigorous unitary matrix construction still requires additional dynamical input, but the hierarchy exponents themselves are uniquely determined by discrete path lengths. $\square$

Corollary 4.6 ($|\rho - i\eta|$ is not an independent parameter). *The third Wolfenstein parameter $|\rho - i\eta|$ is not an independent input but an algebraic consequence of $A = \sqrt{2/3}$ and $V_{ub} = \lambda^3/3$:*

$$|\rho - i\eta| = \frac{V_{ub}}{A\lambda^3} = \frac{\lambda^3/3}{\sqrt{2/3} \cdot \lambda^3} = \frac{1}{3\sqrt{2/3}} = \frac{1}{\sqrt{6}} = 0.408.$$

The experimental value is approximately 0.40, with $\sim 2\%$ deviation.

Remark 4.7 (CKM numerical summary). Complete CKM elements:

$$|V_{us}| = \lambda = 0.222 \text{ (1.1\%)}, \quad |V_{cb}| = A\lambda^2 = 0.040 \text{ (2.3\%)}, \quad |V_{ub}| = \frac{\lambda^3}{3} = 0.00365 \text{ (1.1\%)}. \quad \lambda^3 = 0.00365 \text{ (1.1\%)}$$

All three independent elements deviate by $< 2.5\%$.

4.3 Origin of $A = \sqrt{2/3}$

The Wolfenstein parameter $A \approx 0.81$ controls the magnitude of V_{cb} , i.e., the amplitude coefficient of $b \rightarrow c$ transitions. Physically, A describes the additional suppression of skip-one-generation transitions relative to adjacent-generation transitions.

In the BC framework, A arises from the S_3 permutation symmetry on three-generation space. The three-dimensional permutation representation of S_3 decomposes as

$$\mathbf{3} \cong \mathbf{1} \oplus \mathbf{2}.$$

Here $\mathbf{1}$ corresponds to the common part of three generations (not participating in mixing), while $\mathbf{2}$ carries inter-generational mixing. The normalization coefficient of the projection of $\mathbf{2}$ onto the “adjacent transition channel” in the standard normalized basis is

$$A = \sqrt{\frac{2}{3}} = 0.816.$$

Proposition 4.8 (Representation-Theoretic Origin of A). *In the S_3 permutation representation on three-generation space, $\mathbf{3} = \mathbf{1} \oplus \mathbf{2}$, the normalization projection coefficient of the two-dimensional standard representation is $A = \sqrt{2/3}$.*

All CKM parameters are thus determined by two arithmetic quantities: $\delta_2 = 13.18^\circ$ (Jacobi sum phase) and $\sqrt{2/3}$ (S_3 representation theory). The three independent matrix elements (V_{us} , V_{cb} , V_{ub}) all deviate by $< 2.5\%$.

5 Mass Spectrum

Arithmetic origin of mass. Fermion masses are the most important continuous parameters in particle physics. The nine charged fermion masses span five orders of magnitude—from the electron’s 0.511 MeV to the top quark’s 173 GeV—and the Standard Model does not explain their origin.

In the BC framework, fermion masses can be described by the noncommutativity of addition and multiplication. Each fermion corresponds to a character direction χ_j , and its mass is measured by the commutator $\mu_j^{(a)} = \langle \chi_j | [D_F, T_a] | \chi_j \rangle$, where T_a is the additive translation operator on $\mathbb{Z}/p\mathbb{Z}$ ($T_a|k\rangle = |k+a\rangle$) and D_F is the finite Dirac operator of the BC spectral triple [12, 16, 17, 30]. When $[D_F, T_a] = 0$, addition and multiplication can be separated and the direction is massless; when $[D_F, T_a] \neq 0$, they cannot be separated and the mass is nonzero. The eigenvalues of D_F in the character basis are proportional to the Gauss sum $\tau(\chi_j)$:

$$\mu_j^{(a)} = \langle \chi_j | [D_F, T_a] | \chi_j \rangle. \quad (7)$$

The entanglement of the real character (χ_3) exactly self-cancels at tree level, giving a zero diagonal entry—corresponding to neutrinos; the entanglement of complex characters (χ_2, χ_1) does not self-cancel—corresponding to massive leptons and quarks.

Local seed and global propagation. The commutator $\mu_j^{(a)}$ is a local quantity defined on $\mathbb{Z}/p\mathbb{Z}$. Locally, all nontrivial characters have the same entanglement strength: the Gauss sum satisfies $|\tau(\chi_j)| = \sqrt{p}$ —looking only at $\mathbb{Z}/7\mathbb{Z}$, all fermion masses would be equal, with no hierarchy. The mass hierarchy arises from propagation from local to global: spreading the conflict seed from $\mathbb{Z}/p\mathbb{Z}$ to all of $\mathbb{Z}$, where different characters undergo different modulation at every prime $q = 2, 3, 5, \dots$, yielding the Dirichlet L -functions [23]:

$$L\left(\frac{1}{2}, \chi_j\right) = \sum_{n=1}^{\infty} \frac{\chi_j(n)}{\sqrt{n}}.$$

$s = 1/2$ is the symmetry axis of the functional equation $\Lambda(s, \chi) = \Lambda(1 - s, \bar{\chi})$, and also corresponds to the KMS phase transition point $\beta = 1$ ($s = \beta/2$) of the BC system. $|L(\frac{1}{2}, \chi_j)|$ is no longer equal across different j : $|L_2| = 0.31848$, $L_3 = 1.14659$, $|L_1| = 0.85747$. This unequal modulus is the source of the mass hierarchy—mass differences among fermions arise from different propagation efficiencies of the conflict seed across primes. All fermion masses can therefore be given by the values of $L(\frac{1}{2}, \chi)$ at the critical line.

Fermion masses span 12 orders of magnitude—from the top quark at 173 GeV to neutrinos at ~ 0.05 eV. In the BC framework, all fermion masses originate from different directions of the same character group $(\mathbb{Z}/7\mathbb{Z})^*$, so cross-species ratios are precisely determined by the ratios of L -function values at $s = 1/2$ for different characters:

- Lepton masses are completely determined by the modulus (absolute scale) and phase (three-generation structure) of $L(\frac{1}{2}, \chi_2)$;
- d-type quarks start from leptons, multiplied by the color factor d_2 and the weak-sector weight $L(\frac{1}{2}, \chi_3)/\zeta(\frac{1}{2})$;
- u-type quarks involve all three nontrivial L -functions L_1, L_2, L_3 , reflecting that $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains all characters;
- Neutrinos correspond to the unique real character χ_3 —tree-level mass is zero, and nonzero mass arises from the scattering echo of χ_3 against the χ_0 background (§5.7).

Charged fermions (complex characters) acquire mass through first-order Dirac processes; neutrino mass arises from dilution of the χ_3 signal across N^2 channels in the χ_0 background (§5.7). The 12-order-of-magnitude mass hierarchy is a direct consequence of the real/complex nature of the characters.

All formulas involve four L -function values, each with a clear physical role:

Symbol	Value	Physical role
$L_2 = L(\frac{1}{2}, \chi_2) $	0.31848	Lepton sector scale (modulus $\rightarrow m_\tau$, phase $\rightarrow$ Koide $\rightarrow$ three generations)
$L_3 = L(\frac{1}{2}, \chi_3)$	1.14659	Weak sector weight (d-type cross-species, u-type transition probability)
$L_1 = L(\frac{1}{2}, \chi_1) $	0.85747	u-type sector scale (transition probability)
$\zeta = \zeta(\frac{1}{2}) $	1.46035	Global normalization (trivial character)

Here $L_{01} \equiv L_2$, $L_{10} \equiv L_3$, $L_{11} \equiv L_1$ in CRT double-index notation (subscript ij denotes the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ component).

5.1 Character Paths

The local seed starts from χ_0 (vacuum) and propagates through character space to the sector of each fermion. Which characters the path traverses determines which L -function values appear in the mass formula; the power of each L factor is jointly determined by the Galois structure and propagation coherence of that character.

Power rule. The power of each L factor = Galois degree $\times$ coherence degree:

Galois $\times$ Coherence	Coherent ($C=1$)	Independent ($C=2$)
Complex character ($G=2$)	2	4
Real character ($G=1$)	1	2

Galois degree: complex characters have a Galois pair ($\chi_k \leftrightarrow \bar{\chi}_k$), with modulus squared giving $G = 2$; real characters are self-conjugate, giving $G = 1$. Coherence degree: within the same sector (e.g., χ_2 within leptons) propagation is coherent, $C = 1$; across sectors (e.g., χ_3 from leptons to d-type quarks) propagation is independent Born, $C = 2$. Thus $|L_2|$ has power $2 \times 2 = 4$ in leptons (complex $\times$ independent), L_3 has power $1 \times 1 = 1$ in the b quark (real $\times$ coherent), and power $1 \times 2 = 2$ in s, d quarks (real $\times$ independent).

Lepton: $\chi_0 \rightarrow \chi_2$. Leptons couple only to the even character χ_2 (order 3, $\mathbb{Z}/3\mathbb{Z}$ direction). Single-step propagation; the mass formula involves only L_2 :

$$m_\ell \propto |L_2|^4.$$

Power $4 = G \times C = 2 \times 2$: complex character ($G = 2$), the only nontrivial direction in the lepton sector ($C = 2$, independent Born). This is the shortest path and simplest mass formula.

d-type quark: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$. d-type quarks (b, s, d) additionally traverse χ_3 (order 2, $\mathbb{Z}/2\mathbb{Z}$ direction, weak sector) beyond the lepton path. Two-step propagation; the mass formula involves both L_2 and L_3 :

$$m_d \propto |L_2|^4 \cdot L_3^{\text{power}}.$$

The power of L_3 varies by generation: gen3 (b) has power 1 (coherent, $C = 1$), gen2/gen1 (s, d) have power 2 (independent Born, $C = 2$). The d-type mass hierarchy thus has one additional L_3 modulation beyond leptons.

u-type quark: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$. The u-type quark (c, t, u) character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains all nontrivial character information. Three-step propagation; the mass formula includes all three L values:

$$m_{u\text{-type}} \propto |L_2|^4 \cdot L_3^2 \cdot |L_1|^4.$$

The longest path and most complex formula. $|L_1|^4$ likewise comes from the Born probability of a complex character ($G \times C = 2 \times 2$). u-type quarks “see” the entire $\mathbb{Z}/6\mathbb{Z}$ structure; this is the arithmetic reason why their generation factors involve gauge coupling constants (rather than pure Koide). L factors are the same for all three generations; all mass differences come from generation factors, where the top’s Dyson self-loop and the u quark’s weak coupling will be discussed in the relevant subsections.

Neutrino: $\chi_0 \rightarrow \chi_2 \rightarrow [\chi_0 \text{ roundtrip}] \rightarrow \chi_2$. The neutrino character χ_3 is real ($\chi_3 = \bar{\chi}_3$, order 2). The diagonal commutator of a real character vanishes at tree level: the direct channel is closed. Nonzero mass must be realized through an indirect path.

Character multiplication $\chi_3^2 = \chi_0$ provides the unique indirect channel: the χ_3 signal maps to the trivial character χ_0 via squaring, then returns from χ_0 to χ_3 . Physically, this corresponds to the neutrino field completing a Majorana-type roundtrip via the χ_0 (vacuum/KMS thermal bath) channel—the signal must “borrow” the vacuum channel because its own channel has no propagating modes. χ_0 is the trivial character, carrying $N = \varphi(p^4) = \varphi(p^{\beta-1}) = 2058$ KMS-active modes ($\beta = 5$ is the KMS temperature parameter at which the quark sector freezes in the BC system; $\varphi(p^{\beta-1})$ counts the active

thermal modes at this temperature); the signal is dispersed among N modes upon entry and again upon return.

The L factor doubles due to the roundtrip:

$$m_\nu \propto |L_2|^8.$$

The lepton path gives $|L_2|^4$; the χ_0 roundtrip multiplies by another $|L_2|^4$, yielding $|L_2|^8$.

Path length determines the complexity of the mass formula: more character nodes traversed means more types of L factors and vertex normalization factors. Leptons traverse only χ_2 (one step), while u-type quarks traverse all three nontrivial characters (three steps). The neutrino L factor shares its origin with leptons (both are powers of $|L_2|$), but the N^2 dilution factor from the χ_0 roundtrip produces the 12-order-of-magnitude mass suppression.

5.2 Vertex Factors

Paths determine the L factors in the mass formula. Each fermion mass additionally requires normalization weights at each character node along the path—vertex factors. Vertex factors are determined by the algebraic structure of the CRT decomposition $\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, physically corresponding to the principle: *indistinguishable (symmetric) states are normalized at amplitude level, $\div\sqrt{n}$; distinguishable (independent) states are summed at probability level, $\times n$.*

Upon reaching character χ_k , three independent effects may activate:

1. **Galois projection:** complex characters (χ_2, χ_1) have a Galois pair ($\chi_k \leftrightarrow \bar{\chi}_k$), physically indistinguishable—symmetric state, amplitude-level $\div\sqrt{d_1}$, $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$. Real characters (χ_3) are self-conjugate; no projection, $\times 1$.
2. **Color activation:** χ_3 corresponds to the $\mathbb{Z}/2\mathbb{Z}$ direction of the CRT; three color states are independent in this direction—distinguishable states, probability-level $\times d_2 = \times 3$. Other characters do not activate this.
3. **Background normalization:** when the path crosses from χ_2 to χ_3 , the trivial character χ_0 contribution $\zeta(\frac{1}{2})$ provides the baseline; the crossing cost is $\div|\zeta(\frac{1}{2})|$. Passing through χ_0 itself results in dilution across $N = \varphi(p^4) = 2058$ KMS-active modes, $\div N$ each time.

χ_k	Galois projection	Color activation	Background norm.	Vertex
χ_2	$\div\sqrt{d_1}$	$\times 1$	—	$1/\sqrt{2}$
χ_3	$\times 1$	$\times d_2$	$\div \zeta $	$3/1.46035$
χ_1	$\times 1$	$\times 1$	—	1
χ_0	—	—	$\div N$	$1/2058$

Vertex coefficient application table. Each fermion’s vertex coefficient = the product of single-pass factors at each node $\times$ number of passes:

Particle	$\chi_2 (\div\sqrt{d_1})$	$\chi_3 (\times d_2/ \zeta)$	$\chi_1 (\times 1)$	$\chi_0 (\div N)$	Vertex coeff.
τ, μ, e	$\times 1$	—	—	—	$1/\sqrt{d_1}$
b	$\times 1$	$\times 1$	—	—	$d_2/(\sqrt{d_1} \zeta)$
s, d	$\times 1$	$\times 1$ (Born)	—	—	$1/ \zeta ^2$
c, t, u	$\times 1$	$\times 1$	$\times 1$	—	$1/\sqrt{d_1}$
ν	$\times 2$	—	—	$\times 2$	$d_1^{-1} N^{-2}$

ν passes through χ_2 twice ($(\div\sqrt{d_1})^2 = \div d_1$) and χ_0 twice ($(\div N)^2 = \div N^2$). For s/d , the χ_3 traversal is in independent mode (Born, $C=2$); the color activation d_2 is consumed by the second pass.

The b quark (gen3, the doublet D_1 with greatest tension) shares K_0 with the τ lepton; its cross-sector traversal through χ_3 maintains coherence (amplitude-level propagation, $C=1$)—corresponding to a single color activation $\times d_2$ and single background normalization $\div|\zeta|$. The s, d quarks (gen2/gen1) require additional Koide/Cabibbo rotations for their generation factors; their cross-sector traversal degenerates to independent Born processes ($C=2$)—the color activation factor is absorbed in the Born square, and the vertex simplifies to $1/|\zeta|^2$.

5.3 Unified Mass Formula

Combining paths and vertex factors, each fermion mass is assembled from three factors:

$$m_f = v \times \underbrace{\prod_{\text{path}} |L_k|^{\text{power}}}_{\text{scale}} \times \underbrace{\prod_{\text{vertex}} \text{normalization}}_{\text{vertex factor}} \times \underbrace{\text{Gen}_f}_{\text{generation factor}}. \quad (8)$$

$v = 246.22$ GeV is the experimental Higgs VEV.

Physical meaning of the three factors.

- **Scale** (L factor): Born probability after global propagation of the local conflict seed across $\mathbb{Z}$. The path determines which characters are traversed; the power rule determines the Born level at each step. Mass scale differences across sectors arise entirely from the unequal moduli of $|L_2|, L_3, |L_1|$ at $s = 1/2$.
- **Vertex factor**: normalization weight assigned to each node by the CRT structure of character space. Galois symmetry ($\div\sqrt{d_1}$), color degrees of freedom ($\times d_2$), background baseline ($\div|\zeta|$), and KMS dilution ($\div N$) are all uniquely determined by algebraic structure, with no free parameters.
- **Generation factor** (Gen): mass splitting among three generations within the same sector. Derived from the Jacobi phase δ_2 of §4: leptons and heavy quarks are given by Koide rotation K_k ; the lightest quarks by Cabibbo transition probability λ^2 . The Dyson enhancement and weak coupling for u-type quarks, and Hurwitz weights for neutrinos, are discussed in the relevant subsections.

Among the three factors, the scale and vertex factors are uniquely determined by the path (§5.1) and CRT structure (§5.2); the generation factor is the sole distinction among three generations within a sector. The following subsections expand the unified formula for each of the four fermion types, giving the specific mass for each generation.

5.4 Lepton Masses

The lepton path is $\chi_0 \rightarrow \chi_2$ (§5.1); the unified formula expands to:

$$m_\ell = v \times \underbrace{|L_2|^4}_{\text{L factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{vertex}} \times \underbrace{K_k}_{\text{Gen}}.$$

L factor: $|L_2|^4$. Leptons couple only to the even character χ_2 ; the mass scale is determined by $|L(\frac{1}{2}, \chi_2)|$. $|L_2|^4$ is the Born probability for a complex character ($G \times C = 2 \times 2$, the unique solution of C^* -algebraic positivity and normalization [24]). Dedekind factorization gives its position in the cyclotomic field $K = \mathbb{Q}(\zeta_7)$:

$$y_\tau = |L(\frac{1}{2}, \chi_2)|^4 = \left(\frac{\zeta_{K^+}(\frac{1}{2})}{\zeta(\frac{1}{2})} \right)^2,$$

i.e., the share of the totally real subfield K^+ at the critical line.

Vertex factor: $1/\sqrt{d_1}$. The Galois pair ($\chi_2 \leftrightarrow \bar{\chi}_2$) is physically indistinguishable; symmetric-state amplitude-level normalization gives $\div \sqrt{d_1}$, $d_1 = [\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7)^+] = 2$.

Generation factor: Koide K_k . The three-generation mass ratios are given by the *phase* δ of $L(\frac{1}{2}, \chi_2)$ through the Koide rotation [3, 18]:

$$f_k = 1 + \sqrt{d_1} \cos(\delta + 120^\circ k), \quad K_k = (f_k/f_0)^2.$$

$\delta = -12.73^\circ$ (Jacobi phase δ_2 evaluated at $s^* = \frac{1}{2} + Q/(2\pi)$). $r = \sqrt{d_1} = \sqrt{2}$ yields the Koide formula $Q = 2/3$, valid for any CM field $\mathbb{Q}(\zeta_p)$.

Three-generation mass formulas.

$$m_\tau = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_0 = \frac{|L_2|^4}{\sqrt{2}} \times v, \quad (9)$$

$$m_\mu = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_1 = m_\tau \times (f_1/f_0)^2, \quad (10)$$

$$m_e = v \times |L_2|^4 \times \frac{1}{\sqrt{d_1}} \times K_2 = m_\tau \times (f_2/f_0)^2. \quad (11)$$

$|L_2| = 0.31848$ (scale) and $\delta = -12.73^\circ$ (phase) separately control the absolute mass and inter-generational ratios.

Particle	L factor	Vertex	Gen	Prediction	Experiment	Deviation
τ	$ L_2 ^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777 MeV	+0.8%
μ	$ L_2 ^4$	$1/\sqrt{d_1}$	K_1	106.5 MeV	105.7 MeV	+0.8%
e	$ L_2 ^4$	$1/\sqrt{d_1}$	K_2	0.5140 MeV	0.5110 MeV	+0.7%

All three generations deviate by $< 1\%$. The extreme hierarchy $m_\tau/m_e \approx 3500$ arises from the near-zero crossing $f_2 \rightarrow 0$.

5.5 d-Type Quark Masses

The d-type quark path is $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$ (§5.1), corresponding to the Georgi–Jarlskog relation [4]. The unified formula expands to:

$$m_d = v \times \underbrace{|L_2|^4 \cdot L_3^{\text{pow}}}_{\text{L factor}} \times \underbrace{\text{vertex}}_{\text{see below}} \times \underbrace{\text{Gen}}_{\text{generation}}.$$

L factor: $|L_2|^4 \cdot L_3^{\text{pow}}$. Building on the lepton $|L_2|^4$, d-type quarks additionally traverse χ_3 (weak sector). $L_3 = L(\frac{1}{2}, \chi_3) = 1.14659$ is a real character, self-conjugate ($\chi_3 = \bar{\chi}_3$). The power of L_3 varies by generation: the b quark (gen3) shares the χ_2 direction with leptons; the cross-sector traversal is amplitude-level $\rightarrow L_3^1$; s, d quarks have independent Born for $\chi_3 \rightarrow L_3^2$.

Vertex factor. b quark: path traverses χ_2 (Galois $\div \sqrt{d_1}$) and χ_3 (color activation $\times d_2$, background $\div |\zeta|$); vertex = $d_2/(\sqrt{d_1}|\zeta|) = 3/(\sqrt{2} \times 1.46035)$. s, d quarks: the color activation d_2 at χ_3 is consumed in the second pass (Born level); vertex = $1/|\zeta|^2$.

Generation factor. b (gen3) uses Koide K_0 ; s (gen2) uses Koide K_1 ; d (gen1) uses Koide K_1 times Cabibbo λ^2 . The d quark generation factor is $K_1\lambda^2$: first a Koide rotation to gen2 (K_1), then a Cabibbo hop suppression (λ^2).

Three-generation mass formulas.

$$m_b = v \times |L_2|^4 L_3 \times \frac{d_2}{\sqrt{d_1}|\zeta|} \times K_0, \quad (12)$$

$$m_s = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times K_1, \quad (13)$$

$$m_d = v \times |L_2|^4 L_3^2 \times \frac{1}{|\zeta|^2} \times K_1 \lambda^2. \quad (14)$$

Particle	L factor	Vertex	Gen	Prediction	Experiment	Deviation
<i>b</i>	$ L_2 ^4 L_3$	$d_2/(\sqrt{d_1} \zeta)$	K_0	4219 MeV	4180 MeV	+0.9%
<i>s</i>	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	K_1	92.90 MeV	93.40 MeV	-0.5%
<i>d</i>	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	$K_1 \lambda^2$	4.580 MeV	4.670 MeV	-1.9%

All three generations deviate by $\leq 2\%$. The b/lepton ratio $m_b/m_\tau = d_2 L_3/|\zeta| = 2.356$ (experiment 2.352, deviation +0.1%), i.e., color number times L-function ratio.

5.6 u-Type Quark Masses

The u-type quark path is $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$ (§5.1); the unified formula expands to:

$$m_u = v \times \underbrace{|L_2|^4 \cdot L_3^2 \cdot |L_1|^4}_{\text{L factor}} \times \underbrace{\frac{1}{\sqrt{d_1}}}_{\text{vertex}} \times \underbrace{\text{Gen}}_{\text{generation}}.$$

L factor: $|L_2|^4 L_3^2 |L_1|^4$. The longest path, involving all three L values. Both $|L_2|^4$ and $|L_1|^4$ are Born probabilities of complex characters ($G \times C = 2 \times 2$); L_3^2 is the independent Born of the real character ($G \times C = 1 \times 2$). The u-type character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ contains all nontrivial characters, so u-type quarks “see” the entire $\mathbb{Z}/6\mathbb{Z}$ structure. Unlike the amplitude-level L_3^1 for d-type quarks, u-type quarks have a different character ($\chi_1 \neq \chi_2$) from leptons, and the cross-sector factor degenerates to Born probability level.

Vertex factor: $1/\sqrt{d_1}$. The path traverses χ_2 (Galois $\div \sqrt{d_1}$), χ_3 , and χ_1 . The χ_3 and χ_1 vertices are transparent (§5.2); total vertex factor = $1/\sqrt{d_1} = 1/\sqrt{2}$, same as leptons.

Generation factor. The u-type character $\chi_1 = \bar{\chi}_2 \cdot \chi_3$ simultaneously contains the electromagnetic direction (χ_2) and weak direction (χ_3); hence its generation factor is no longer a pure Koide rotation but is modulated by coupling constants of the character directions along the path—the unifying principle is **which character-direction couplings are activated along the path**:

- c (gen2, reference generation): Koide K_0 , no coupling constant activated—baseline;
- t (gen3): the self-loop in the χ_2 direction ($\chi_2^2 = \bar{\chi}_2$) allows an infinite series of virtual EM scattering; Born series summation gives Dyson enhancement $1/\alpha_{\text{EM}} \approx 137$;
- u (gen1): each inter-generation hop through the χ_3 direction incurs a weak-force cost, Cabibbo λ^2 times $\alpha_W = \alpha_{\text{EM}}/\sin^2\theta_W \approx 1/29.6$ (θ_W is the Weinberg angle, $\sin^2\theta_W = 0.2312$).

Note: $K_0 = 1$ marks the **reference generation with no additional inter-generational modulation** in each sector, not a fixed assignment to gen3. In leptons and d-type quarks, gen3 happens to be the heaviest with no additional coupling, so the reference generation = gen3; in u-type quarks, gen3 (t) is shifted from the Koide baseline by Dyson enhancement, making gen2 (c) the reference generation with no additional coupling.

Three-generation mass formulas.

$$m_c = v \times |L_2|^4 |L_3^2| L_1|^4 \times \frac{1}{\sqrt{d_1}} \times K_0, \quad (15)$$

$$m_t = v \times |L_2|^4 |L_3^2| L_1|^4 \times \frac{1}{\sqrt{d_1}} \times \frac{1}{\alpha_{\text{EM}}}, \quad (16)$$

$$m_u = v \times |L_2|^4 |L_3^2| L_1|^4 \times \frac{1}{\sqrt{d_1}} \times \lambda^2 \alpha_W. \quad (17)$$

L factors and vertices are identical across all three generations; all differences reside in the generation factor.

Particle	L factor	Vertex	Gen	Prediction	Experiment	Deviation
c	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	K_0	1273 MeV	1270 MeV	+0.2%
t	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$1/\alpha_{\text{EM}}$	174.5 GeV	172.8 GeV	+1.0%
u	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$\lambda^2 \alpha_W$	2.119 MeV	2.160 MeV	-1.9%

All three generations deviate by $\leq 2\%$. $m_t/m_c = 1/\alpha_{\text{EM}} \approx 137$: the anomalously large top mass is a direct consequence of the EM self-loop. $m_u/m_c = \lambda^2 \alpha_W \approx 1/600$: the extreme lightness of the u quark results from Cabibbo suppression $\times$ weak-force suppression.

5.7 Neutrino Masses

The neutrino path is $\chi_0 \rightarrow \chi_2 \rightarrow [\chi_0 \text{ roundtrip}] \rightarrow \chi_2$ (§5.1); the unified formula expands to:

$$m_\nu = v \times \underbrace{|L_2|^8}_{\text{L factor}} \times \underbrace{\frac{1}{d_1 N^2}}_{\text{vertex}} \times \underbrace{\lambda_k}_{\text{Gen}}.$$

L factor: $|L_2|^8$. The χ_0 roundtrip doubles the lepton path's $|L_2|^4$. Physically, the neutrino character χ_3 is real ($\chi_3 = \bar{\chi}_3$); the direct channel is closed. $\chi_3^2 = \chi_0$ provides the unique indirect channel: the signal takes a roundtrip through χ_0 , equivalent to traversing the lepton path twice: $|L_2|^4 \times |L_2|^4 = |L_2|^8$.

Vertex factor: $1/(d_1 N^2)$. χ_2 is traversed twice: Galois projection $(\div \sqrt{d_1})^2 = \div d_1$. χ_0 is traversed twice: KMS dilution $(\div N)^2 = \div N^2$, $N = \varphi(p^4) = 2058$. Total vertex $= 1/(d_1 N^2)$. This is the primary reason for the extreme smallness of neutrino masses: dilution across $N^2 \approx 4.2 \times 10^6$ channels.

Generation factor: Hurwitz weights λ_k . Neutrino generations are not given by Koide K_k but by eigenvalues of the Koide–Hurwitz mass matrix (unlike the traditional seesaw mechanism [10, 11, 21, 28], no high-energy scale M_R is introduced here).

Doublet products $V_k = a_k(p - a_k)$ satisfy the Fermat condition $V_k^3 \equiv -1 \pmod{7}$ — $-1 = \text{involution} = \text{the arithmetic expression of the Majorana condition}$; $n = 3 = (p - 1)/2$ is the minimal unifying power.

The Hurwitz ζ -function lifts the local Fermat structure to the global level:

$$w_k = \frac{1}{p^6} \zeta_H\left(3, \frac{a_k}{p}\right) \times \zeta_H\left(3, \frac{p - a_k}{p}\right).$$

The Koide circulant matrix $C = \text{circ}(f_0, f_2, f_1)$ combined with Hurwitz weights gives the mass matrix:

$$\mathcal{M} = C \cdot \text{diag}(w_1, w_2, w_3) \cdot C^T.$$

C is shared with charged leptons (same Koide parameters $r = \sqrt{2}$, phase δ), but $w_1 \neq w_2 \neq w_3$ breaks Koide’s $\mathbb{Z}/3\mathbb{Z}$ cyclic symmetry, causing eigenvalues λ_k to deviate from the simple K_k .

Three-generation mass formulas.

$$m_k = v \times |L_2|^8 \times \frac{1}{d_1 N^2} \times \lambda_k, \quad k = 1, 2, 3. \quad (18)$$

Numerical values: $\lambda_1 = 1.660$, $\lambda_2 = 3.250$, $\lambda_3 = 16.45$; scale $v|L_2|^8/(d_1 N^2) = 3.080$ meV.

Particle	L factor	Vertex	Gen	Prediction	Experiment	Deviation
ν_3	$ L_2 ^8$	$1/(d_1 N^2)$	λ_3	50.67 meV	~ 50 meV	+1.3%
ν_2	$ L_2 ^8$	$1/(d_1 N^2)$	λ_2	10.01 meV	—	—
ν_1	$ L_2 ^8$	$1/(d_1 N^2)$	λ_1	5.113 meV	—	—

$\Sigma m_\nu = 65.79$ meV, within the CMB-S4 sensitivity (~ 30 meV) [25]. $R = \Delta m_{31}^2/\Delta m_{21}^2 = 34.31$, consistent with experiment 34.0 ± 1.4 [9] (0.2σ).

CP phase. The CP-violating phase of neutrinos is likewise determined by the arithmetic properties of χ_3 . The Gauss sum $\tau(\chi_3) = i\sqrt{7}$ is purely imaginary; its argument locks $\delta_{CP} = -\arg(\tau(\chi_3)) = -\pi/2 = -90^\circ$, consistent with the NOvA/T2K best fit [7, 8].

6 Discussion and Predictions

Sections 2–5 have demonstrated that 12 fermion masses are completely determined by the character structure, Jacobi phase, and Dirichlet L -function values at the critical line of the BC system, with all deviations $\leq 2\%$.

6.1 Summary Table

The following summarizes all fermion mass predictions. Each mass is given by the unified formula (§5.3): $m_f = v \times |L|^{\text{pow}} \times \text{vertex} \times \text{Gen}$.

Particle	Path	L factor	Vertex	Gen	Prediction	Experiment	Deviation
τ	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_0	1791 MeV	1777	+0.8%
μ	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_1	106.5	105.7	+0.8%
e	$\chi_0 \rightarrow \chi_2$	$ L_2 ^4$	$1/\sqrt{d_1}$	K_2	0.514	0.511	+0.7%
b	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3$	$d_2/(\sqrt{d_1} \zeta)$	K_0	4219	4180	+0.9%
s	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	K_1	92.90	93.40	-0.5%
d	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3$	$ L_2 ^4 L_3^2$	$1/ \zeta ^2$	$K_1 \lambda^2$	4.58	4.67	-1.9%
c	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	K_0	1273	1270	+0.2%
t	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$1/\alpha_{\text{EM}}$	174500	172800	+1.0%
u	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$	$ L_2 ^4 L_3^2 L_1 ^4$	$1/\sqrt{d_1}$	$\lambda^2 \alpha_W$	2.119	2.160	-1.9%
ν_3	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_0 \rightarrow \chi_2$	$ L_2 ^8$	$1/(d_1 N^2)$	λ_3	50.67 meV	~ 50	+1.3%
ν_2	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_0 \rightarrow \chi_2$	$ L_2 ^8$	$1/(d_1 N^2)$	λ_2	10.01 meV	—	—
ν_1	$\chi_0 \rightarrow \chi_2 \rightarrow \chi_0 \rightarrow \chi_2$	$ L_2 ^8$	$1/(d_1 N^2)$	λ_1	5.113 meV	—	—

All nine charged fermions deviate by $\leq 2\%$; neutrino $\Sigma m_\nu = 65.79 \text{ meV}$, $R = 34.31$. All derived from four L -function values $+N = \varphi(p^4)$; v , α_{EM} , α_W are experimental inputs.

6.2 Testable Predictions

(1) Neutrinos. This paper provides independent absolute mass predictions for three neutrino generations:

$$m_1 = 5.113 \text{ meV}, \quad m_2 = 10.01 \text{ meV}, \quad m_3 = 50.67 \text{ meV}.$$

m_1 and m_2 have not been directly measured; their detection would constitute direct verification. Other neutrino predictions:

- $\Sigma m_\nu = 65.79 \text{ meV}$: CMB-S4 [25] sensitivity $\sim 30 \text{ meV}$ can directly test this;
- $R = \Delta m_{31}^2 / \Delta m_{21}^2 = 34.31$: independent of the phase δ and absolute scale, this is the most central numerical prediction of this paper. JUNO [13] will measure to $< 1\%$ precision; current $R_{\text{exp}} = 34.0 \pm 1.4$ [9] (0.2σ);
- $\delta_{CP} = -90^\circ$: locked by the Gauss sum $\tau(\chi_3) = i\sqrt{7}$, not fitted. DUNE [29] and Hyper-Kamiokande [14] will measure to $\pm 5^\circ$;
- Normal ordering (NH): $m_3 > m_2 > m_1$. If JUNO confirms inverted ordering, the framework is falsified;
- Majorana type: $\chi_3 = \bar{\chi}_3$ (self-conjugate) requires neutrinos to be Majorana particles. $0\nu\beta\beta$ experiments (nEXO/LEGEND) can test this.

(2) Exact equalities between mass ratios and coupling constants. The three u-type quark generations share the same L factors and vertex factors; mass ratios are entirely determined by generation factors. This yields two exact equalities:

$$\frac{m_t}{m_c} = \frac{1}{\alpha_{\text{EM}}(0)} = 137.036, \quad (19)$$

$$\frac{m_u}{m_c} = \lambda^2 \alpha_W. \quad (20)$$

Equation (19) binds the top/charm mass ratio to the electromagnetic coupling constant: currently $m_t/m_c = 172800/1270 = 136.1$, deviation 0.7%. The HL-LHC and future e^+e^- colliders will push top and charm mass precision to the 0.1% level; convergence of m_t/m_c to 137.036 would constitute strong validation, while deviation would falsify the prediction. Equation (20) similarly binds the u/charm ratio to the Cabibbo angle and weak coupling.

(3) CKM mixing parameters. Three independent CKM elements are uniquely determined by δ_2 and S_3 representation theory:

$$|V_{us}| = 0.222 \text{ (1.1\%)}, \quad |V_{cb}| = 0.040 \text{ (2.3\%)}, \quad |V_{ub}| = 0.00365 \text{ (1.1\%)}.$$

LHCb and Belle II are pushing precision to the 0.1% level.

(4) Higgs mass. The three disjoint doublets D_1, D_2, D_3 of $(\mathbb{Z}/7\mathbb{Z})^*$ are each split by the involution ι into a symmetric state ($\iota = +1$, scalar) and an antisymmetric state ($\iota = -1$, fermion), each with probability 1/2. The Higgs field, being scalar, couples only to the symmetric branch within each doublet. The three doublets are independent; the joint probability of the scalar branches is:

$$\lambda_H = \left(\frac{1}{2}\right)^3 = \frac{1}{8}, \quad m_H = v\sqrt{2\lambda_H} = \frac{v}{2} = 123.1 \text{ GeV}.$$

Experiment gives 125.10 GeV, deviation -1.6% .

(5) Structural predictions.

- **Strict three generations:** arising from the rigid structure of $\mathbb{Z}/3\mathbb{Z}$. Discovery of a fourth fundamental fermion generation would falsify the framework.

6.3 Open Problems

Although this paper has assembled most of the storyline, several key open problems must be honestly stated:

- **RH/critical line problem:** This paper repeatedly uses critical-line phase locking and finite-character rigidity, suggesting that Riemann-type spectral structure may participate at a deeper level, but a complete connection remains to be established;
- **Origin of $p = 7$:** Why does nature select precisely 7, rather than another prime? This paper takes 7 as the minimal viable input, but its ultimate origin may require a higher-level selection principle or an absolute geometry explanation.

6.4 Conclusion

The 12 fermion masses of the Standard Model—from the top quark at 173 GeV to neutrinos at ~ 0.05 eV, spanning 12 orders of magnitude—were previously treated as independent experimental inputs. This paper provides the unified formula:

$$\boxed{m_f = v \times |L|^{\text{pow}} \times \text{vertex} \times \text{Gen},} \quad (21)$$

reducing 12 masses to four Dirichlet L -function values at the critical line $s = 1/2$. All nine charged fermion deviations are $\leq 2\%$; neutrino masses ($m_1 = 5.113$, $m_2 = 10.01$,

$m_3 = 50.67 \text{ meV}$, $\Sigma m_\nu = 65.79 \text{ meV}$) and mass ratio $R = 34.31$ are derived from the same framework. The exact equalities $m_t/m_c = 1/\alpha_{\text{EM}}$ and $m_u/m_c = \lambda^2\alpha_W$ further bind the mass spectrum to force constants.

JUNO (2026–2027) measurement of R to $< 1\%$, CMB-S4 sensitivity of $\sim 30 \text{ meV}$ for Σm_ν , and HL-LHC precision measurements of m_t/m_c and the Higgs self-coupling λ will constitute direct tests of this framework.

References

- [1] J.-B. Bost and A. Connes, “Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory,” *Selecta Math.* **1** (1995) 411–457.
- [2] A. Connes, *Noncommutative Geometry*, Academic Press, 1994.
- [3] Y. Koide, “A new view of quark and lepton mass hierarchy,” *Phys. Rev. D* **28** (1983) 252–254.
- [4] H. Georgi and C. Jarlskog, “A new lepton-quark mass relation in a unified theory,” *Phys. Lett. B* **86** (1979) 297–300.
- [5] L. Wolfenstein, “Parametrization of the Kobayashi-Maskawa matrix,” *Phys. Rev. Lett.* **51** (1983) 1945–1947.
- [6] Particle Data Group, “Review of Particle Physics,” 2024 edition.
- [7] M. A. Acero *et al.* (NOvA Collaboration), “Improved measurement of neutrino oscillation parameters by the NOvA experiment,” *Phys. Rev. D* **106** (2022) 032004.
- [8] K. Abe *et al.* (T2K Collaboration), “Measurements of neutrino oscillation parameters from the T2K experiment using 3.6×10^{21} protons on target,” *Eur. Phys. J. C* **83** (2023) 782.
- [9] I. Esteban *et al.*, “NuFIT 6.0: updated global analysis of three-flavour neutrino oscillations,” *JHEP* **09** (2024) 178.
- [10] P. Minkowski, “ $\mu \rightarrow e\gamma$ at a rate of one out of 10^9 muon decays?” *Phys. Lett. B* **67** (1977) 421–428.
- [11] T. Yanagida, “Horizontal symmetry and masses of neutrinos,” in *Proceedings of the Workshop on Unified Theory and Baryon Number in the Universe*, ed. O. Sawada and A. Sugamoto, KEK, 1979, pp. 95–99.
- [12] A. Connes and M. Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*, American Mathematical Society, 2008.
- [13] F. An *et al.* (JUNO Collaboration), “JUNO physics and detector,” *Prog. Part. Nucl. Phys.* **123** (2022) 103927.
- [14] K. Abe *et al.* (Hyper-Kamiokande Collaboration), “Hyper-Kamiokande design report,” arXiv:1805.04163.

- [15] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer, 1990.
- [16] A. H. Chamseddine, A. Connes, and M. Marcolli, “Gravity and the standard model with neutrino mixing,” *Adv. Theor. Math. Phys.* **11** (2007) 991.
- [17] A. H. Chamseddine and A. Connes, “The spectral action principle,” *Commun. Math. Phys.* **186** (1997) 731.
- [18] Y. Koide, “New view of quark and lepton mass hierarchy,” *Phys. Rev. D* **71** (2005) 016010.
- [19] N. Cabibbo, “Unitary symmetry and leptonic decays,” *Phys. Rev. Lett.* **10** (1963) 531.
- [20] M. Kobayashi and T. Maskawa, “ CP -violation in the renormalizable theory of weak interaction,” *Prog. Theor. Phys.* **49** (1973) 652.
- [21] M. Gell-Mann, P. Ramond, and R. Slansky, “Complex spinors and unified theories,” in *Supergravity* (P. van Nieuwenhuizen and D. Freedman, eds.), North-Holland, 1979, p. 315.
- [22] C. D. Froggatt and H. B. Nielsen, “Hierarchy of quark masses, Cabibbo angles and CP violation,” *Nucl. Phys. B* **147** (1979) 277.
- [23] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer, 2000.
- [24] W. H. Zurek, “Probabilities from entanglement, Born’s rule $p_k = |\psi_k|^2$ from enviance,” *Phys. Rev. A* **71** (2005) 052105.
- [25] CMB-S4 Collaboration, “CMB-S4 science case, reference design, and project plan,” arXiv:2203.08024.
- [26] B. C. Berndt, R. J. Evans, and K. S. Williams, *Gauss and Jacobi Sums*, Wiley, 1998.
- [27] O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2*, 2nd ed., Springer, 1997.
- [28] R. N. Mohapatra and G. Senjanović, “Neutrino mass and spontaneous parity non-conservation,” *Phys. Rev. Lett.* **44** (1980) 912.
- [29] B. Abi *et al.* (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), far detector technical design report, volume II,” arXiv:2002.03005.
- [30] A. H. Chamseddine and A. Connes, “Why the standard model,” *J. Geom. Phys.* **58** (2008) 38–47.